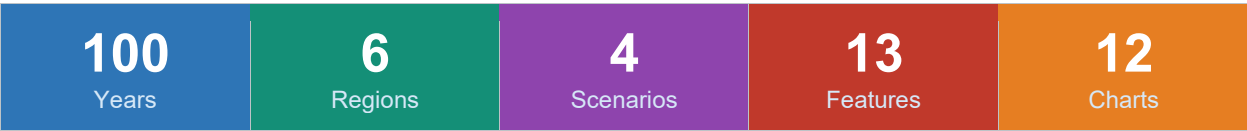


EXPLORATORY DATA ANALYSIS • FULL REPORT

Global Climate Change Dataset

Historical Records + Future Projections | 1940 – 2040

Python • Pandas • Seaborn • Matplotlib • SciPy



1. Project Overview

This report presents a comprehensive Exploratory Data Analysis (EDA) of the Global Climate Change Extended Dataset spanning 1940 to 2040. Combining historical climate observations with model-based future projections, it offers a long-range perspective on how environmental systems have transformed and where current trajectories may lead.

The analysis addresses twelve core research questions through statistical summaries, correlation analysis, and twelve targeted visualizations — all produced in Python using Pandas, Seaborn, Matplotlib, and SciPy.

Property	Details
File	climate_dataset_extended_1940_2040.csv
Year Range	1940 – 2040 (100 years of records)
Scenarios	Historical • Current (2020–2026) • BAU • Green
Regions	North America • South America • Europe • Asia • Africa • Oceania
Features	13 climate & socio-economic columns
Format	CSV / UTF-8
Visualizations	12 charts — bar, scatter, line, box, heatmap, clustermap

2. Dataset & Feature Descriptions

Column	Type	Description	Unit
Year	Integer	Year of observation	1940–2040
Country	String	Country name	Text
Region	String	Geographic region grouping	Text
Scenario	String	Climate trajectory scenario	Hist/BAU/Green/Current
CO2_Emissions_mt	Float	Carbon dioxide emissions	Metric tons
Temp_Anomaly_C	Float	Temperature deviation from baseline	°C
Renewable_Energy_%	Float	Share of renewables in total energy mix	%
Extreme_Events	Integer	Count of extreme climate events	Count
Sea_Level_mm	Float	Sea level rise from 1940 baseline	mm
Methane_ppb	Float	Atmospheric methane concentration	ppb
GDP_Billion	Float	National GDP	USD Billion
Climate_Policy_Index	Float	Composite policy strength score	Index 0–100
Population_Million	Float	National population	Millions

3. Libraries & Data Preprocessing

3.1 Python Libraries

```
import pandas as pd          # Data loading, cleaning, aggregation
import numpy as np           # Numerical operations
import matplotlib.pyplot as plt # Base plotting
import seaborn as sns        # Statistical visualizations
from scipy.stats import skew, kurtosis # Distribution analysis
```

Pandas served as the primary data handling library, enabling efficient loading of the CSV dataset, cleaning of missing values, column-level aggregations, and multi-dimensional groupby operations across regions and scenarios.

NumPy provided the numerical computation foundation underlying all mathematical operations — including array manipulation, statistical calculations, and matrix-level transformations used internally by both Pandas and Seaborn.

Matplotlib (pyplot) functioned as the base visualization engine, offering granular control over chart elements including axes, labels, tick marks, color schemes, grid lines, and figure sizing across all twelve visualizations in this report.

Seaborn was layered on top of Matplotlib to produce statistically rich, publication-quality charts with minimal code — including bar plots, scatter plots, regression plots, box plots, heatmaps, and the hierarchical clustermap used in the correlation analysis.

SciPy — Skewness quantified the directional lean of the CO₂ emission distribution — a positive value confirmed a right-skewed tail driven by industrialization-era emission spikes that pull the mean above the median.

SciPy — Kurtosis measured tail heaviness relative to a normal distribution, confirming that extreme emission outliers occur with greater frequency than standard models predict — a critical consideration before applying any machine learning pipeline to this dataset.

3.2 Loading & Inspection

```
df = pd.read_csv('climate_dataset_extended_1940_2040.csv')
df.shape          # Rows x columns
df.head()         # First 5 rows
df.info()         # Data types + null counts
df.describe()     # Statistical summary
df.isnull().sum() # Missing value check
df.sample(5)      # Random sanity-check sample
df.memory_usage(deep=True)
```

The dataset was loaded and systematically inspected using a structured sequence of Pandas operations to establish a complete understanding of its structure, quality, and memory profile before any analysis was performed.

pd.read_csv() ingested the raw CSV file into a Pandas DataFrame — a two-dimensional, labeled table structure that supports filtering, aggregation, and transformation operations across all 13 climate and socio-economic columns.

df.shape returned the exact dimensions of the dataset as a (rows, columns) tuple, confirming the total number of observations across the 100-year span and the breadth of features available for analysis.

df.head() displayed the first five rows of the DataFrame, providing an immediate visual confirmation that the data loaded correctly, column names were intact, and values appeared within expected ranges.

df.info() produced a column-level summary including data types (integer, float, object) and non-null counts for every feature — the first diagnostic step for identifying which columns contained missing values and whether type inference was applied correctly during loading.

df.describe() generated a full statistical summary for all numeric columns, returning count, mean, standard deviation, minimum, maximum, and the 25th, 50th, and 75th percentiles — establishing the baseline distribution profile for every quantitative variable in the dataset.

df.isnull().sum() counted the total number of missing values in each column, flagging any gaps in early historical years (1940–1960) where global climate monitoring infrastructure was sparse and incomplete records are expected.

df.sample(5) returned five randomly selected rows from across the full dataset, serving as a quick sanity check to verify that data from different years, regions, and scenarios was loaded consistently and without corruption.

df.memory_usage(deep=True) reported the precise RAM consumption of each column in bytes, providing essential information for optimizing performance when scaling the analysis to larger datasets or deploying models in memory-constrained environments.

3.3 Categorical Exploration & Aggregations

```
df['Region'].value_counts()
df['Scenario'].unique()
df['Country'].nunique()
df.corr(numeric_only=True)
df.groupby('Region')['CO2_Emissions_mt'].mean()
df.groupby('Region')[['CO2_Emissions_mt', 'Temp_Anomaly_C']].mean()
```

`df['Region'].value_counts()` → Counts how many times each region appears in the dataset and shows the frequency of records per region.

`df['Scenario'].unique()` → Displays all unique scenario names/categories present in the Scenario column without duplicates.

`df['Country'].nunique()` → Returns the total number of distinct countries available in the dataset.

`df.corr(numeric_only=True)` → Calculates the correlation matrix for all numeric columns to show relationships between variables (values range from -1 to 1).

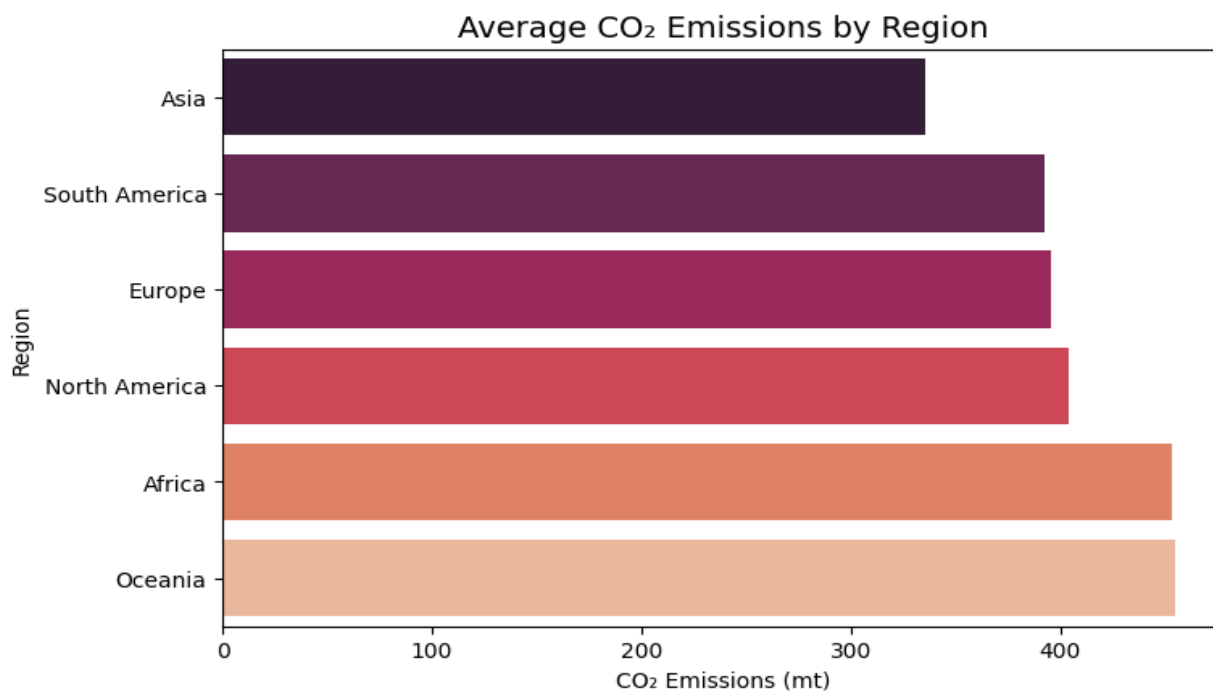
`df.groupby('Region')[['CO2_Emissions_mt', 'Temp_Anomaly_C']].mean()` → Groups data by region and computes the average CO2 emissions and temperature anomaly for each region.

4. Visualizations & Findings

4.1. Average CO₂ Emissions by Region

Question: Which regions contribute the highest CO₂ emissions?

```
region_co2 = df.groupby('Region')['CO2_Emissions_mt'].mean().sort_values()
palette = sns.color_palette('rocket', len(region_co2))
sns.barplot(x=region_co2.values, y=region_co2.index,
            hue=region_co2.index, palette=palette, legend=False)
plt.title('Average CO2 Emissions by Region')
```

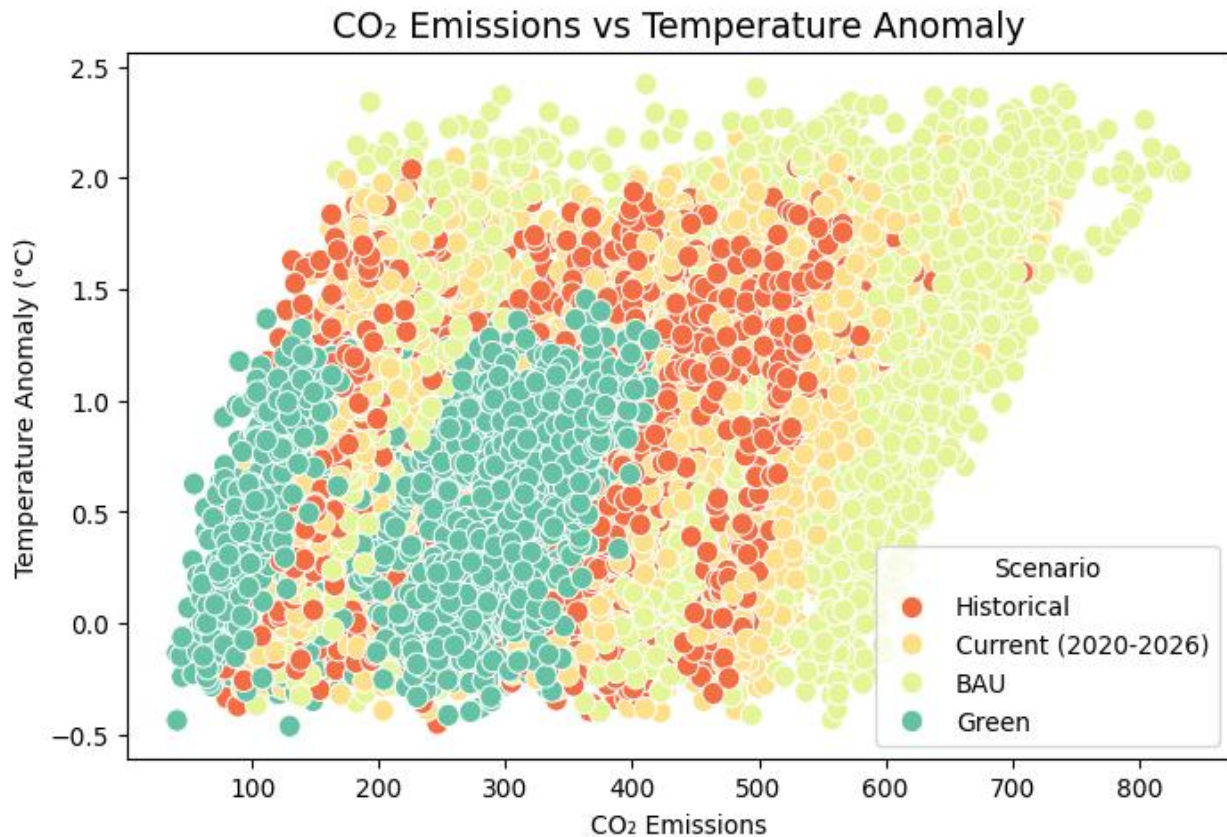


#	Insight	Implication
1	Oceania & Africa lead at ~450 mt — significantly above all other regions	Primary policy targets for emission reduction programs
2	Asia is the lowest emitter (~330 mt) despite being the most populous region	Population size does not determine emission levels
3	Europe & South America are nearly equal at ~390 mt	Similar emission profiles despite contrasting economic structures
4	North America sits mid-table at ~405 mt	Moderate-high output; strong policy leverage available here
5	~120 mt gap separates highest and lowest emitting regions	Regional disparity demands differentiated climate policy

4.2.CO₂ Emissions vs Temperature Anomaly

Question: *Is temperature anomaly increasing with CO₂ emissions?*

```
sns.scatterplot(data=df, x='CO2_Emissions_mt', y='Temp_Anomaly_C',  
                hue='Scenario', palette='Spectral', s=80)
```

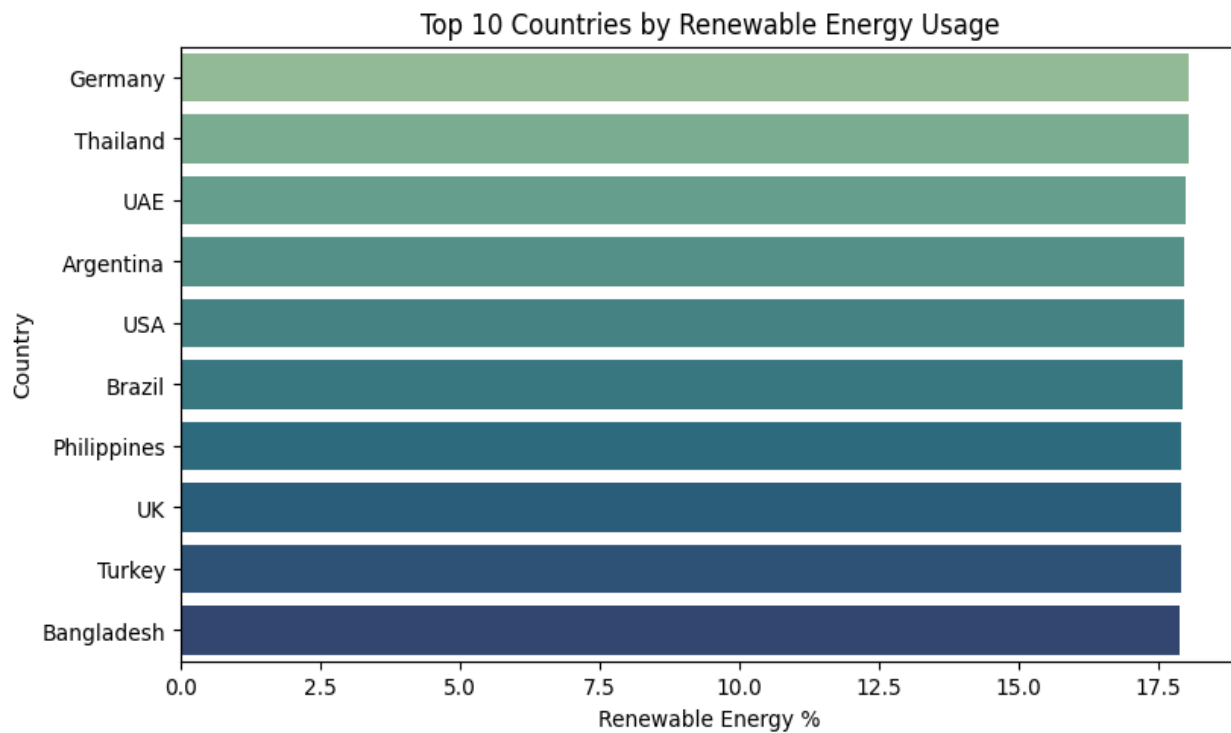


#	Insight	Implication
1	Green scenario clusters between -0.5°C and +1.5°C — the lowest anomaly band	Aggressive emission reduction directly suppresses temperature rise
2	BAU extends to 800+ CO ₂ and +2.5°C — the most alarming projection	Inaction exponentially expands the emission and temperature ceiling
3	Positive correlation exists across all four scenarios	Higher CO ₂ consistently aligns with higher temperature anomalies
4	Current (2020–2026) heavily overlaps with Historical data	Emission trajectories have not shifted meaningfully from historical patterns
5	BAU emission range widens far beyond all other scenarios	The gap between action and inaction grows dramatically over time

4.3. Top 10 Countries by Renewable Energy Usage

Question: Which countries invest most in renewable energy?

```
top = df.groupby('Country')['Renewable_Energy_%'].mean()
      .sort_values(ascending=False).head(10)
sns.barplot(x=top.values, y=top.index, hue=top.index,
            palette='crest', legend=False)
```

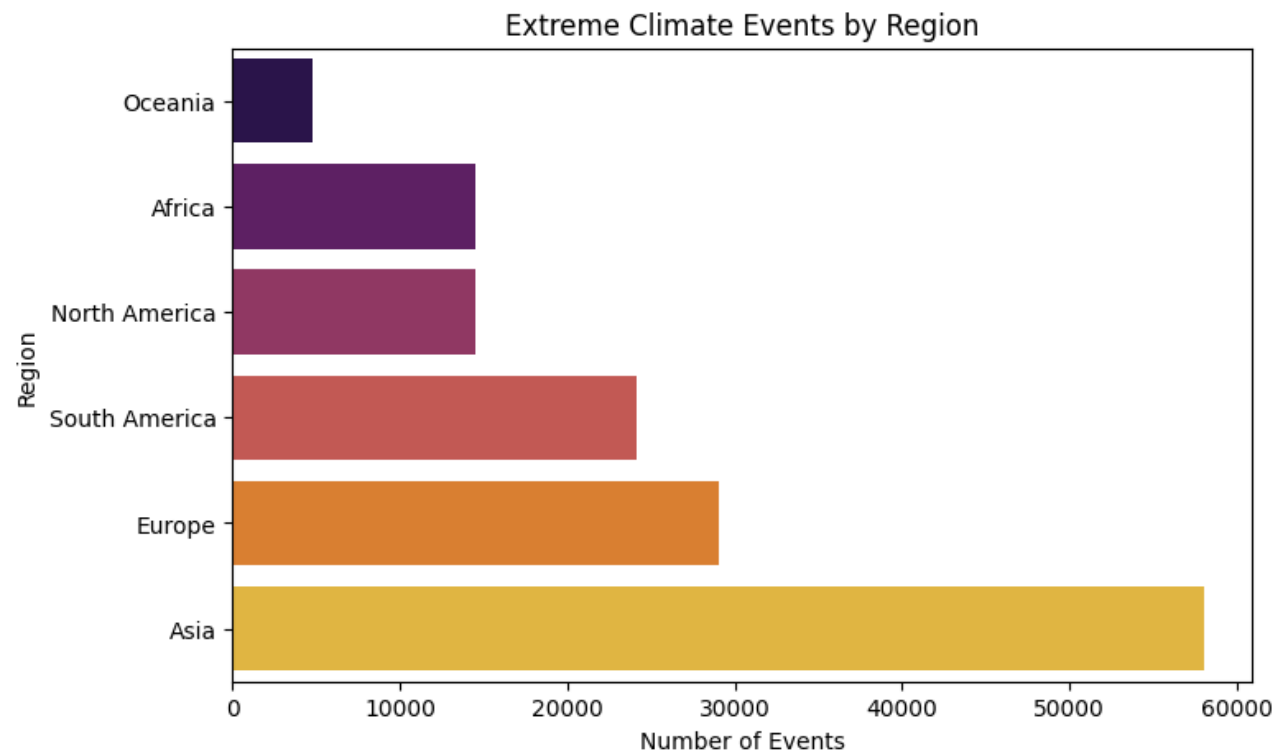


#	Insight	Implication
1	All 10 countries sit at ~18% renewable energy — differences are negligibly small	A global renewable adoption plateau exists across diverse economies
2	Germany leads the chart, reflecting its Energiewende energy transition policy	National policy commitment is the top driver of renewable adoption
3	Developed and developing nations share identical percentages (UAE, Bangladesh)	Renewable adoption is not solely a function of economic development level
4	UAE's presence signals a significant strategic shift away from fossil fuels	Even fossil-fuel-dominant economies are diversifying toward clean energy
5	Bangladesh achieving ~18% is remarkable for a lower-income developing nation	International financing and technology transfer are likely key enablers

4.4. Extreme Climate Events by Region

Question: Which regions face the most extreme climate events?

```
events = df.groupby('Region')['Extreme_Events'].sum().sort_values()
sns.barplot(x=events.values, y=events.index,
            hue=events.index, palette='inferno', legend=False)
```

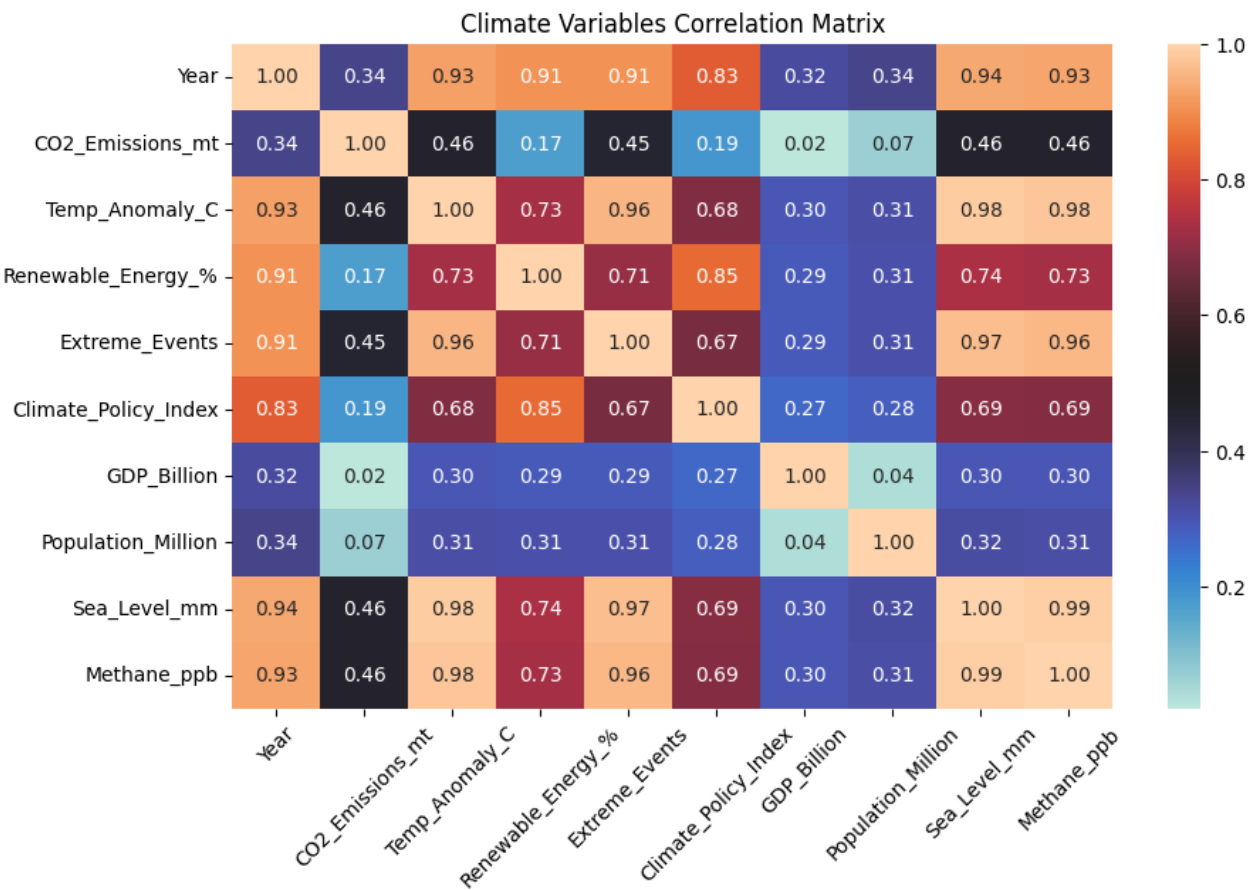


#	Insight	Implication
1	Asia dominates with ~58,000 extreme events — nearly double Europe’s count	Asia is the most climate-vulnerable region on the planet by a wide margin
2	Europe is second with ~29,000 events — a surprisingly high frequency	Escalating heatwaves, floods, and wildfires are driving European exposure
3	Africa & North America are nearly identical at ~14,000 events each	Comparable exposure despite very different geographic and economic profiles
4	Oceania records the fewest events at ~5,000	Smaller landmass and lower population density explain the lower absolute count
5	Asia–Oceania gap exceeds 11x — an extreme regional disparity	Regional equity in climate financing must reflect this imbalance directly

4.5.Climate Variables Correlation Matrix

Question: How strongly are climate variables correlated with each other?

```
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap='icefire', fmt='.2f')
plt.xticks(rotation=45)
```

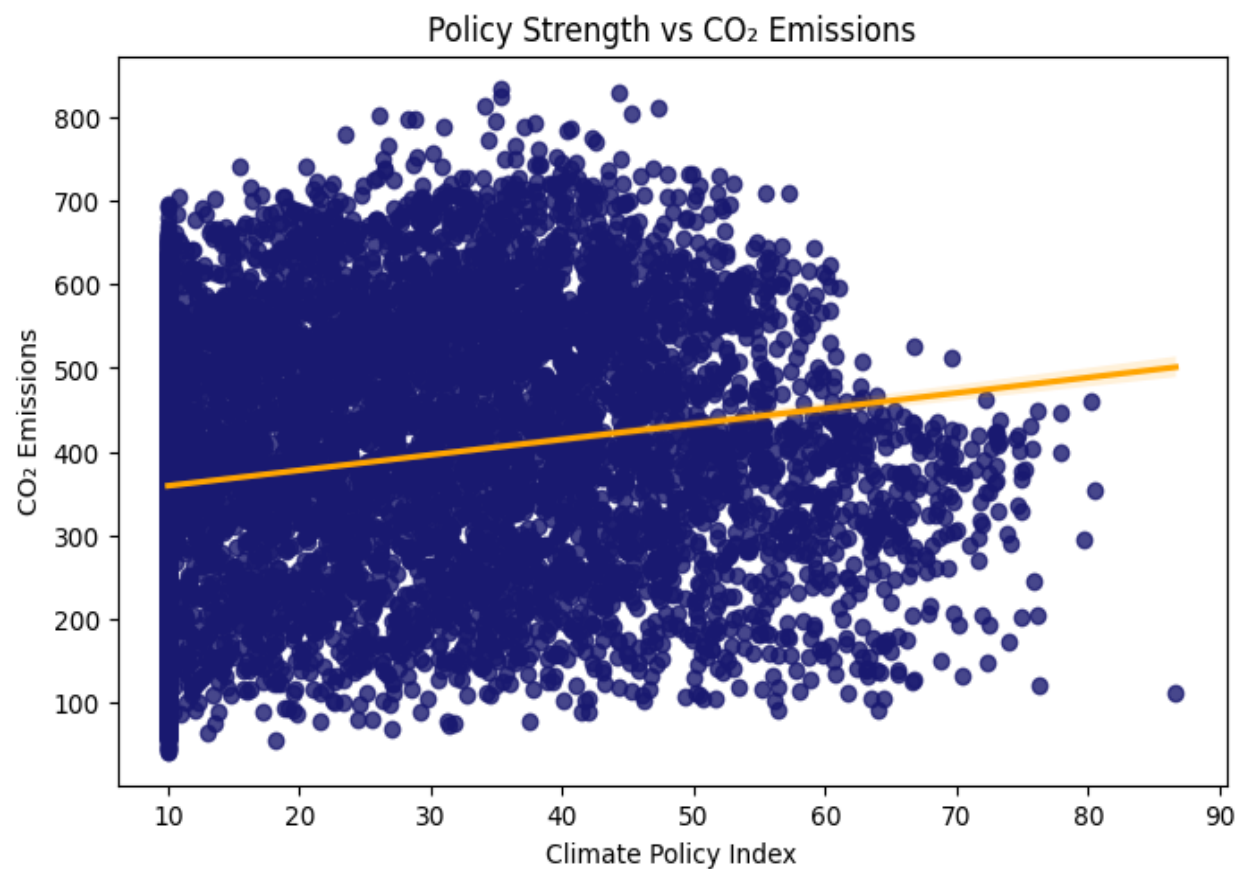


#	Insight	Implication
1	Sea Level & Methane are near-perfectly correlated (0.99)	Rising methane is a direct contributor to accelerating sea level rise
2	Temp_Anomaly_C drives almost everything (0.96–0.98 with Sea Level & Methane)	Temperature anomaly is the central connecting variable across all indicators
3	CO ₂ Emissions correlate weakly with GDP (0.02) and Population (0.07)	Emissions are structural, not explained by economic or demographic size
4	GDP & Population show uniformly low climate correlations (0.02–0.32)	Wealthier or larger nations do not automatically produce worse outcomes
5	Renewable Energy & Climate Policy are strongly linked (0.85)	Stronger policies directly translate into greater renewable adoption

4.6. Policy Strength vs CO₂ Emissions

Question: Are stronger climate policies reducing emissions?

```
sns.regplot(data=df, x='Climate_Policy_Index', y='CO2_Emissions_mt',
            scatter_kws={'color':'midnightblue'}, line_kws={'color':'orange'})
```

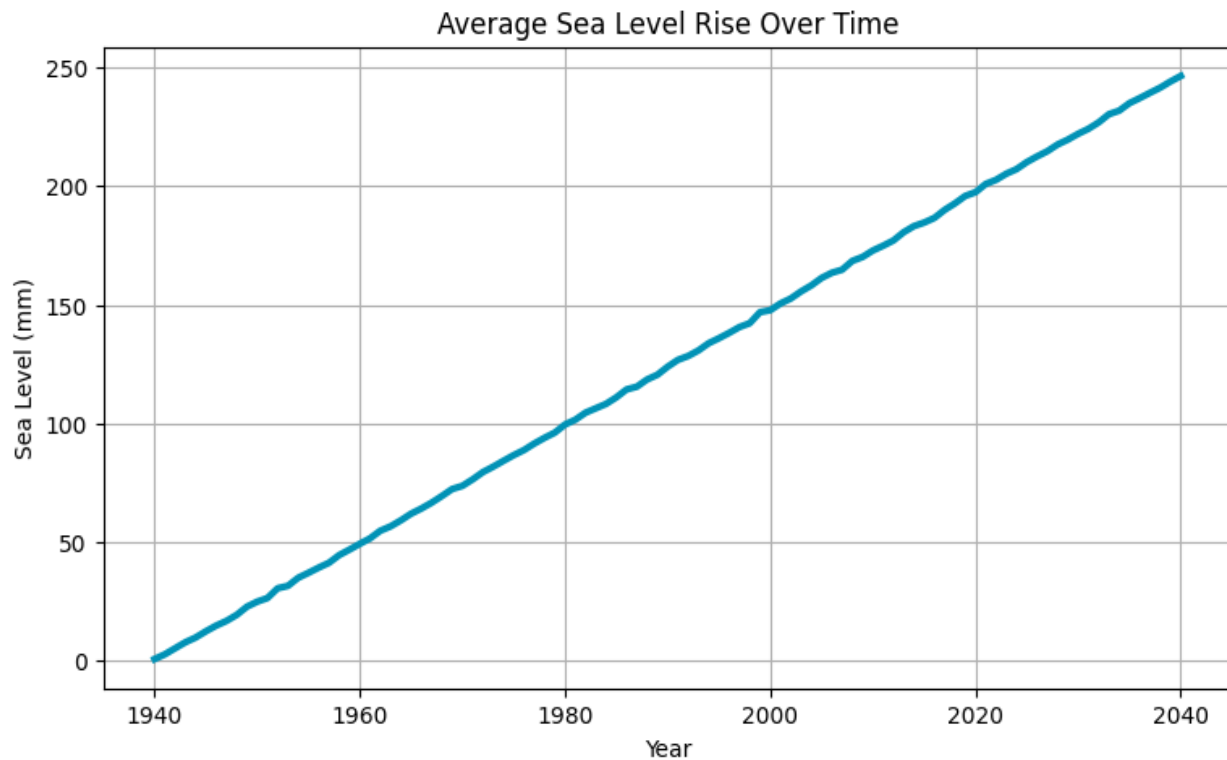


#	Insight	Implication
1	Regression line shows a counterintuitive positive slope	Policy lag effects delay measurable emission reductions
2	High emission outliers (700–800 CO ₂) exist at every policy index level	Policy strength alone is insufficient — enforcement matters too
3	Weak policies (10–30) produce the most volatile emission ranges	Low-policy environments are the most unpredictable and risky
4	Very strong policies (60+) show a tighter band around 300–500 CO ₂	High-commitment regimes begin to stabilize output over time
5	Low emissions appear at both extremes — a non-linear relationship	Policy-emission dynamics are complex and context-dependent

4.7. Average Sea Level Rise Over Time

Question: How has sea level changed over the full 100-year period?

```
sea = df.groupby('Year')['Sea_Level_mm'].mean()
plt.plot(sea.index, sea.values, linewidth=3)
plt.grid(True)
```

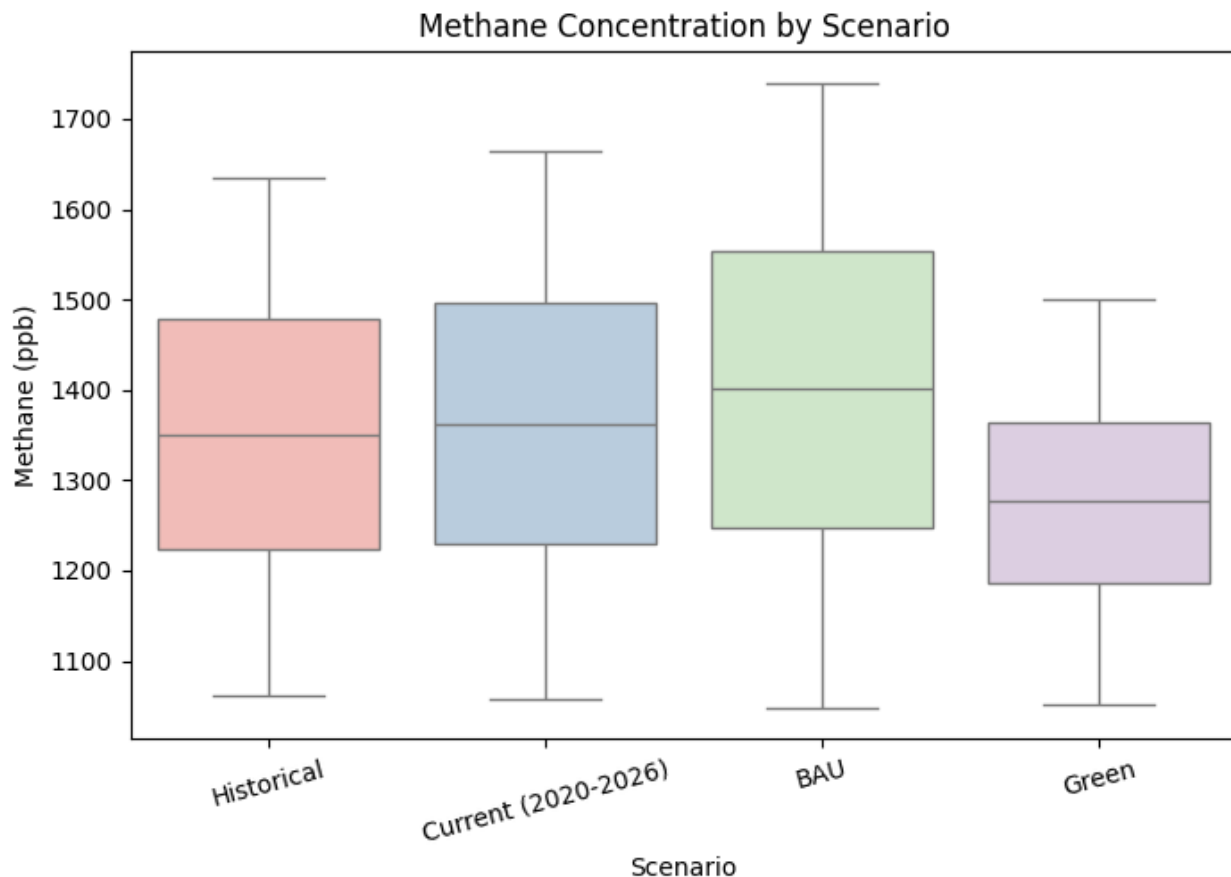


#	Insight	Implication
1	Sea levels rose linearly from 0 mm (1940) to ~248 mm (2040) — zero slowdown	The rise is consistent, relentless, and shows no sign of reversing
2	Rate averages ~2.5 mm per year across the entire century	Small annual gains compound into massive long-term consequences
3	No policy intervention is reflected anywhere in the trend line	Decades of climate agreements have not bent the sea level curve
4	Post-2020 projection adds ~50 mm in just 20 years	The final segment is the most consequential of the entire dataset
5	Total ~248 mm (25 cm) threatens coastal cities and island nations globally	Storm surges, flooding, and displacement will intensify sharply

4.8. Methane Concentration by Scenario

Question: Which scenario produces the highest methane concentration?

```
sns.boxplot(data=df, x='Scenario', y='Methane_ppb',  
            hue='Scenario', palette='Pastell', legend=False)  
plt.xticks(rotation=15)
```

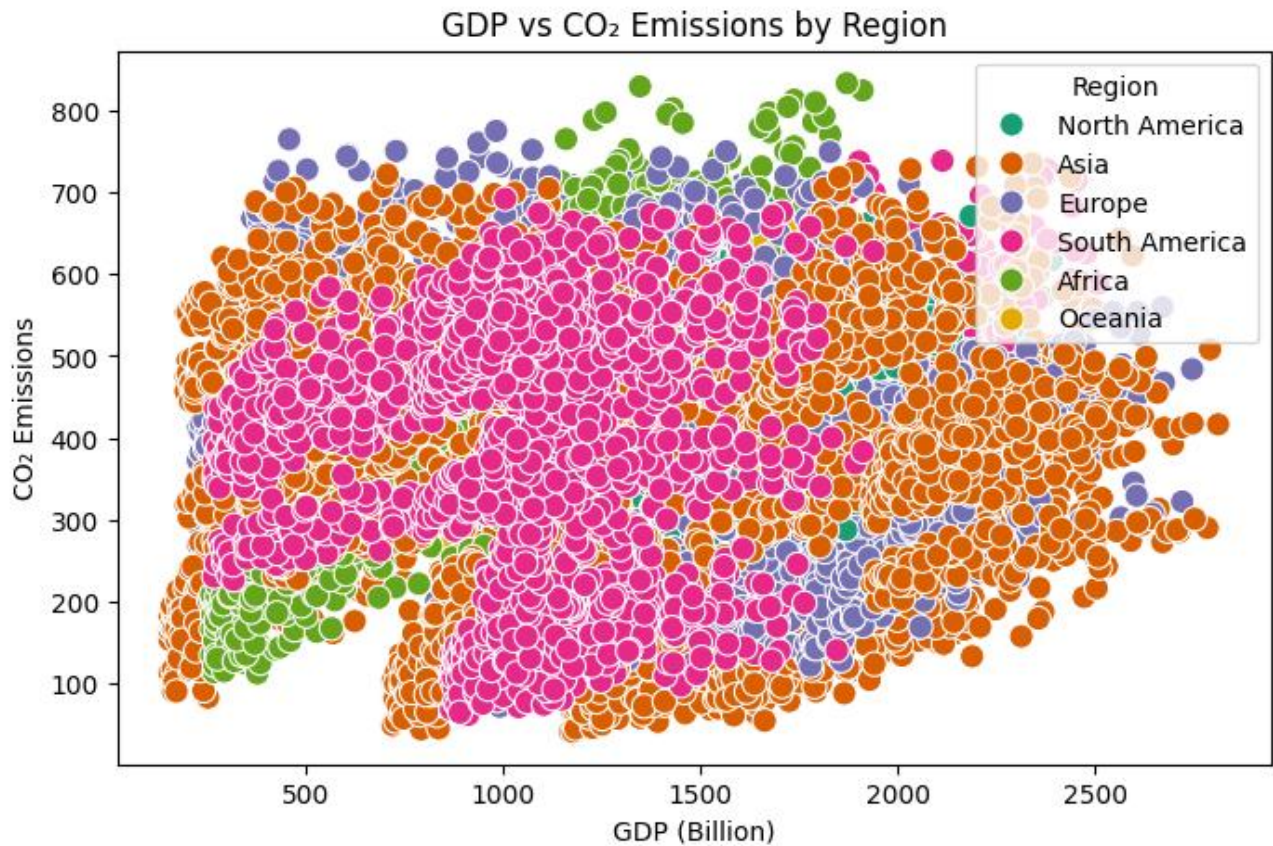


#	Insight	Implication
1	BAU has the highest median methane (~1400 ppb) reaching ~1700 ppb peak	Inaction produces the most severe atmospheric methane outcomes
2	Green scenario shows the lowest median (~1275 ppb) and tightest IQR	Policy reduces both the level and variability of methane simultaneously
3	Current (2020–2026) closely mirrors historical methane patterns	No meaningful departure from historical trends has occurred yet
4	BAU carries the widest uncertainty band — inaction creates volatility	Higher methane also means less predictable future conditions
5	Green scenario has the smallest spread of all four scenarios	Strong policy creates stability, not just reduction, in atmospheric levels

4.9.GDP vs CO₂ Emissions by Region

Question: Which regions have the highest GDP and emissions together?

```
sns.scatterplot(data=df, x='GDP_Billion', y='CO2_Emissions_mt',
                hue='Region', palette='Dark2', s=90)
```

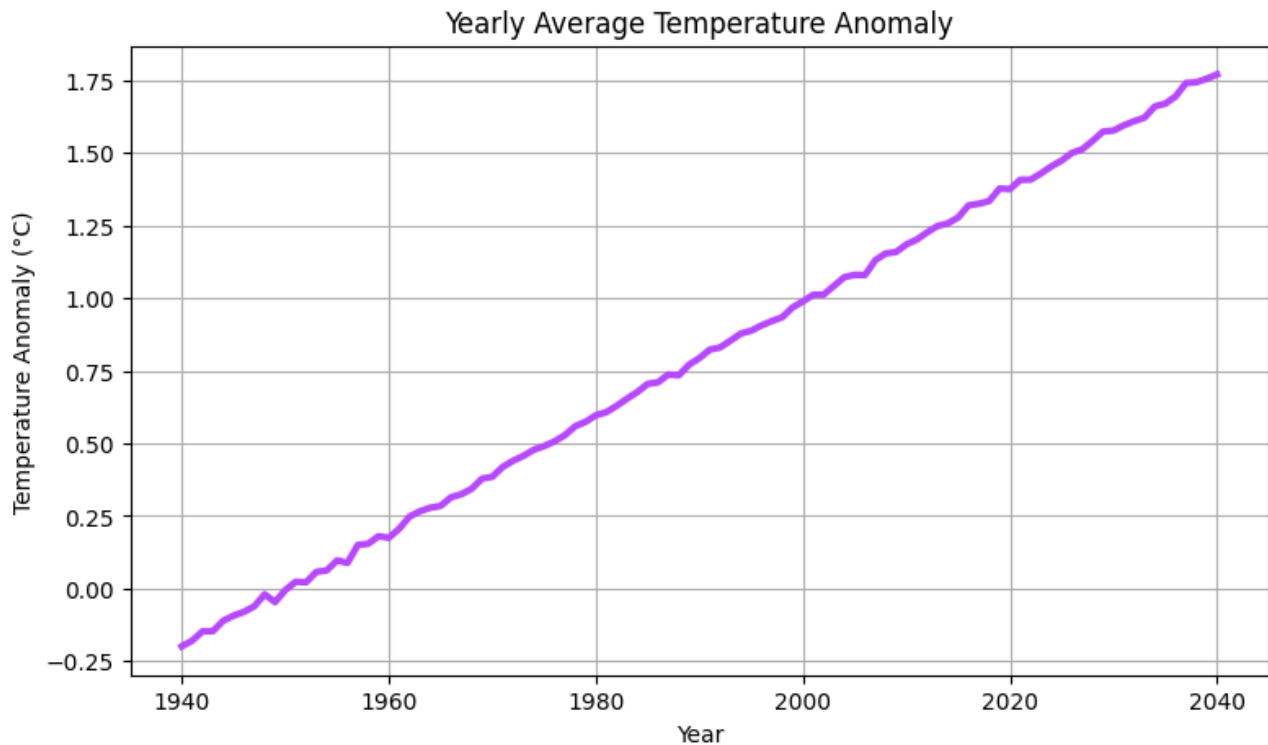


#	Insight	Implication
1	No clear linear relationship exists between GDP and CO ₂ emissions	Wealthy nations are not necessarily the worst emitters
2	Asia dominates emission intensity across the full GDP spectrum	Asia's emissions are structural, not purely wealth-driven
3	North America produces the highest individual emission peaks (700–800+)	Fewer records but extreme outliers at the absolute top
4	South America clusters in the mid-GDP, mid-emission zone consistently	Relatively uniform economic and emission profile continent-wide
5	Africa shows low GDP but widely spread emissions	Energy mix and industrialization type drive the emission dispersion

4.10. Yearly Average Temperature Anomaly

Research Question: Which years recorded the highest average temperature anomaly?

```
temp = df.groupby('Year')['Temp_Anomaly_C'].mean()
palette = sns.color_palette('cool')
sns.lineplot(x=temp.index, y=temp.values, color=palette[4], linewidth=3)
plt.grid(True)
```

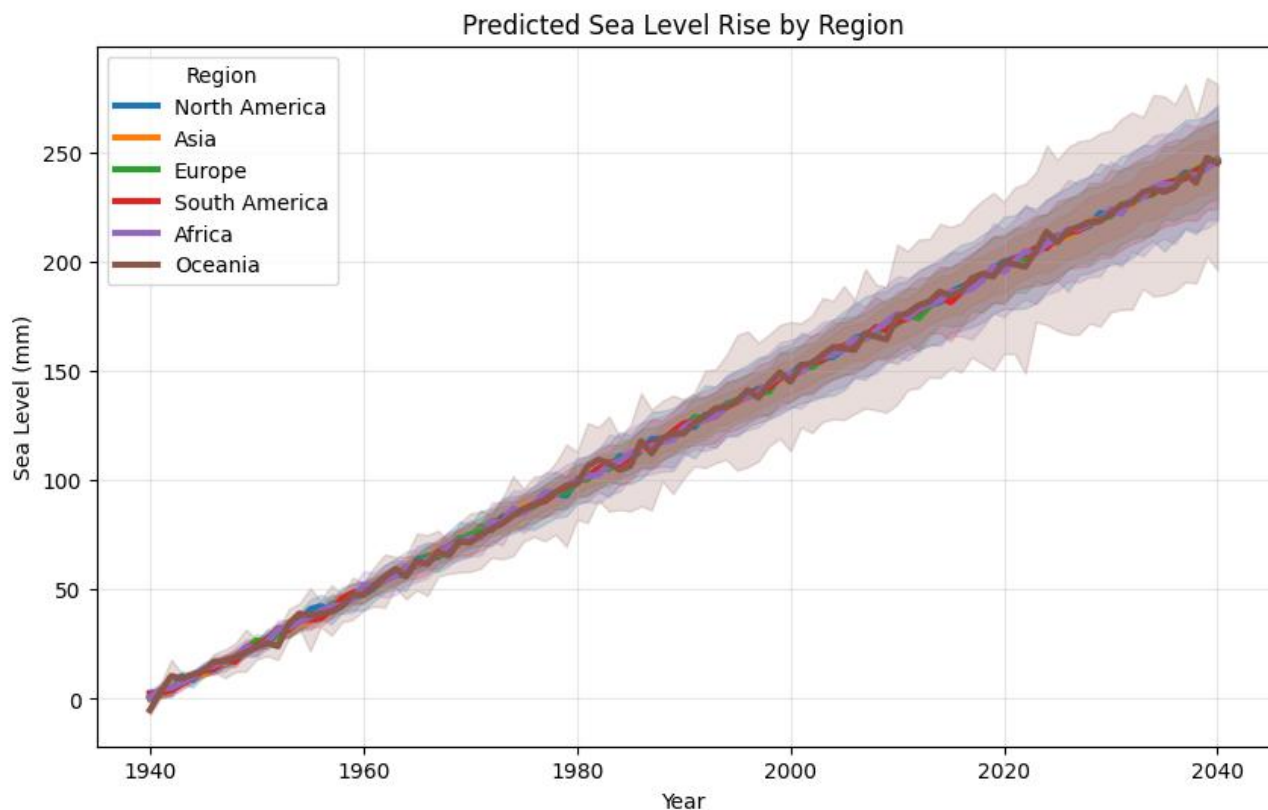


#	Insight	Implication
1	Temperature rose ~2°C in 100 years — from -0.27°C (1940) to +1.77°C (2040)	Unprecedented warming rate in Earth's recorded climate history
2	1.5°C Paris Agreement threshold is breached around 2030–2035	The agreed 'safe' warming limit will be surpassed this decade
3	Warming visibly accelerates post-1980 with global industrial expansion	Human activity has dramatically intensified the warming trajectory
4	Pre-1960 represents the last era of relative temperature stability	A narrow historical window of climate equilibrium has permanently closed
5	Post-2020 shows no plateau — on track to exceed +2°C before 2050	Without major intervention, two degrees will become the new floor

4.11. Predicted Sea Level Rise by Region

Question: Which regions are likely to experience the most severe sea level rise?

```
sns.lineplot(data=df, x='Year', y='Sea_Level_mm',  
             hue='Region', palette='tab10', linewidth=3)  
plt.grid(alpha=0.3)
```

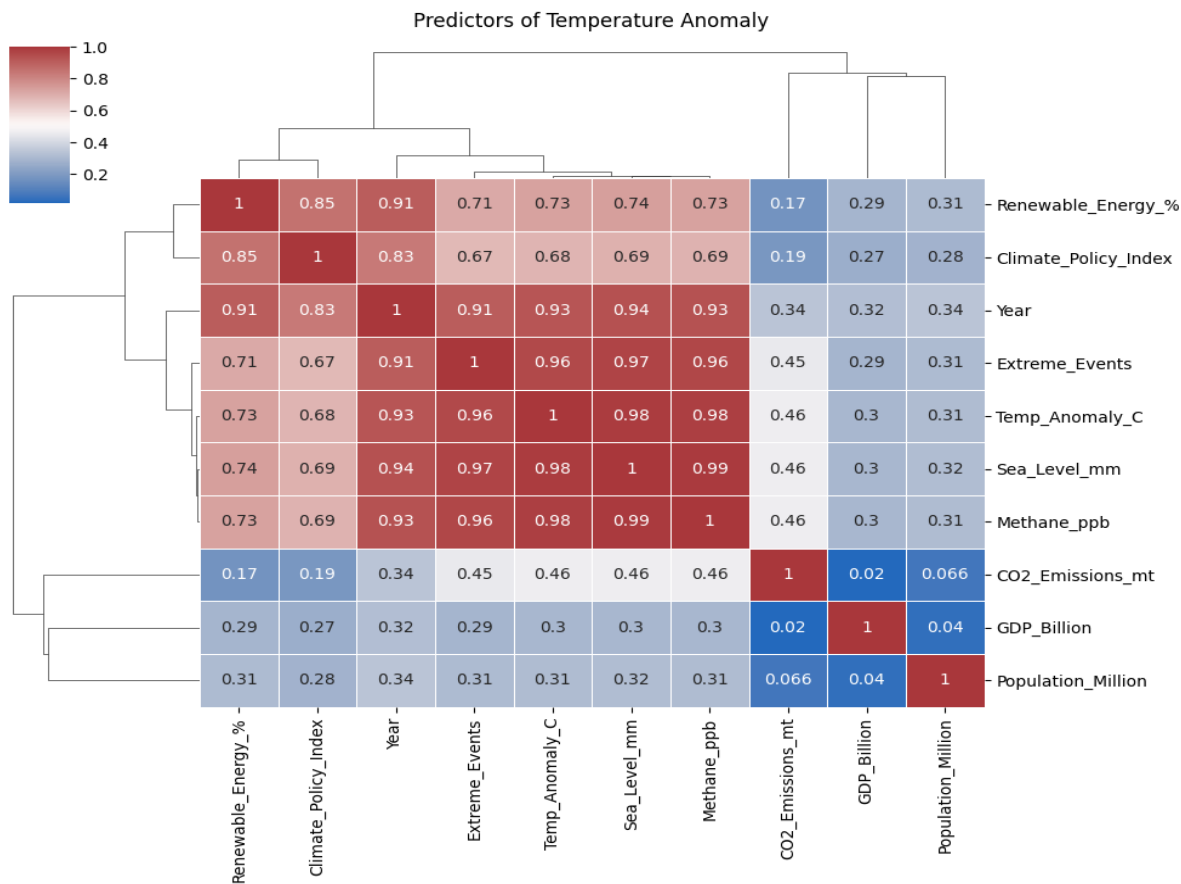


#	Insight	Implication
1	All six regions show near-identical trajectories — sea level rise is globally uniform	No region is immune; all coastal areas require adaptation planning
2	Uncertainty bands widen drastically post-2000 across all regions	Future projections become increasingly uncertain over time
3	Oceania carries the widest uncertainty band, reaching ~280 mm by 2040	Island nations face the greatest variability in climate risk exposure
4	Growth rate visibly accelerates post-1980 — mirrors temperature acceleration	Sea level rise is directly linked to the same warming drivers
5	All regions converge at ~245 mm by 2040, threatening coastal infrastructure	A 25 cm rise will trigger cascading coastal and societal impacts globally

4.12. Predictors of Temperature Anomaly (Clustermap)

Research Question: Which variables most strongly predict temperature anomaly?

```
corr = df.corr(numeric_only=True)
sns.clustermap(corr, cmap='vlag', annot=True,
               figsize=(10,8), linewidths=0.5)
```



#	Insight	Implication
1	Temp Anomaly, Sea Level & Methane form a near-perfect cluster (0.98–0.99)	A self-reinforcing climate feedback loop is actively operating
2	Renewable Energy & Climate Policy cluster together (0.85)	They are inseparable twin levers that must advance simultaneously
3	CO ₂ Emissions stand isolated from all core climate variables	Emissions are driven by structural, not demographic, factors
4	GDP & Population are climate-irrelevant in predictive terms	National wealth and size are poor footprint/vulnerability proxies
5	Year (0.91–0.94) is a master proxy for all climate deterioration	Time itself is the strongest predictor — conditions worsen every year

5. Advanced Statistical Analysis

5.1 Multi-Dimensional Aggregation

```
result = df.groupby(['Region', 'Scenario']).agg(  
    CO2_Avg      = ('CO2_Emissions_mt', 'mean'),  
    Temp_Avg     = ('Temp_Anomaly_C', 'mean'),  
    Events_Total = ('Extreme_Events', 'sum')  
) .reset_index()
```

5.2 Pivot Table — CO₂ by Region vs Scenario

```
pivot = df.pivot_table(  
    values='CO2_Emissions_mt', index='Region',  
    columns='Scenario', aggfunc='mean')
```

5.3 Cross-Tabulation — Scenario Frequency per Region

```
cross = pd.crosstab(df['Region'], df['Scenario'])
```

5.4 Distribution Shape Analysis

```
from scipy.stats import skew, kurtosis  
print('Skewness :', skew(df['CO2_Emissions_mt']))  
# Positive skew = right tail; industrialization-era emission spikes  
print('Kurtosis :', kurtosis(df['CO2_Emissions_mt']))  
# High kurtosis = heavy tails; extreme emission events more frequent than normal
```

Statistic	What It Measures	CO ₂ Interpretation
Skewness	Symmetry of the distribution	Positive skew indicates industrialization-era emission spikes pulling the right tail
Kurtosis	Tail heaviness relative to normal distribution	High kurtosis means extreme emission outliers occur more often than expected

5.5 Sorting & Filtering

```
# Sort by highest emissions  
df_sorted = df.sort_values(by='CO2_Emissions_mt', ascending=False)  
  
# Multi-key sort: region alphabetically, then emissions descending  
df.sort_values(by=['Region', 'CO2_Emissions_mt'], ascending=[True, False])  
  
# Filter records above critical thresholds  
df.query("CO2_Emissions_mt > 500 and Temp_Anomaly_C > 1.5")
```

6. Key Takeaways

The following findings represent the most critical, evidence-backed conclusions drawn from the full century of climate data. Each finding is paired with its direct implication for action.

6.1 Temperature & Emissions

#	Finding	Action Implication
1	Temperature rose ~2°C in 100 years — Paris 1.5°C threshold breached ~2030–2035	Immediate, deep decarbonisation is unavoidable
2	BAU is dramatically worse across every indicator — methane, CO ₂ , temperature	The cost of inaction compounds exponentially over time
3	Green scenario delivers the best outcomes for every climate variable	Policy-driven clean energy transition demonstrably works
4	CO ₂ emissions are NOT driven by GDP or population — structural factors dominate	Sectoral and energy-mix reforms are the critical levers

6.2 Sea Level, Methane & Extreme Events

#	Finding	Action Implication
5	Sea levels rising ~2.5 mm/year with zero slowdown across all regions	Coastal adaptation infrastructure must begin now
6	Methane & sea level are near-perfectly correlated (0.99) — a feedback loop	Methane reduction is as urgent and critical as CO ₂ reduction
7	Asia bears 11× more extreme events than Oceania	Regional climate equity must be central to financing decisions
8	Renewable energy and climate policy must advance together ($r = 0.85$)	Integrated, simultaneous policy design is non-negotiable

7. Next Steps & Recommendations

Based on the patterns, correlations, and anomalies identified throughout this EDA, the following steps are recommended for researchers, data scientists, policymakers, and institutions working with this dataset or related climate data.

7.1 Analytical Next Steps

Recommended Action	Rationale & Expected Outcome
Build time-series forecasting models (ARIMA, LSTM, Prophet)	Year correlates 0.91–0.94 with all major climate indicators — time-based models will have strong predictive power
Develop scenario classification models	The four scenarios (Historical, Current, BAU, Green) are analytically distinct and suitable for ML classification tasks
Apply IQR / z-score filtering before modelling	Emission spikes in early industrialization years create outliers that will distort regression and clustering algorithms
Impute or interpolate missing values in 1940–1960	Sparse early monitoring data should be handled explicitly before any ML pipeline to avoid biased model training
Conduct region-specific sub-analyses	Asia's extreme event dominance and Oceania's sea level variability suggest region-level models will outperform global ones
Explore non-linear models for policy-emission dynamics	The counterintuitive positive slope in Policy vs CO ₂ suggests linear regression alone will not capture the true relationship

7.2 Policy Recommendations

Pri.	Recommendation	Supporting Evidence
 HIGH	Prioritise Green scenario policies globally without delay	BAU vs Green gap widens every decade — delayed action is permanently more costly
 HIGH	Address methane emissions alongside CO₂ in all climate frameworks	Methane-sea level correlation (0.99) means methane reduction directly slows sea level rise
 MED	Direct disproportionate climate financing toward Asia	Asia faces 11× the extreme events of Oceania — current financing allocations do not reflect this exposure
 MED	Advance renewable energy and climate policy simultaneously	Decoupling these ($r = 0.85$) risks policy effectiveness — they must be co-designed
 STD	Begin mandatory coastal adaptation planning for all low-elevation regions	Sea level rise is linear and uninterrupted — 25 cm by 2040 is now a planning baseline, not a projection
 STD	Reform emission accounting to focus on structural energy-mix factors	CO ₂ is not correlated with GDP or population, meaning per-capita or GDP-weighted targets miss the real drivers

7.3 Dataset Extension Opportunities

Extension	Value Added
Add country-level granularity within regions	Regional averages mask high-variance within-region dynamics, especially in Asia and Africa
Incorporate Arctic/Antarctic ice extent data	Ice extent is a critical indicator missing from current features; it directly drives sea level acceleration
Include energy consumption mix by source	Distinguishing coal, gas, nuclear, and renewables per country would make the energy-policy relationship far more actionable
Link to economic shock events (2008, COVID-19)	Emission dips during economic shocks are visible in data but unexplained without event markers
Add carbon capture and sequestration metrics	Green scenario modelling would be strengthened by accounting for removal alongside reduction

8. Disclaimer & Citation

Important Disclaimer

Future values (2026–2040) are model-based simulated projections created for research and educational purposes only. They are NOT official scientific forecasts from IPCC, NASA, NOAA, or any governmental body. Always cross-reference with authoritative sources before making policy or academic decisions.

How to Cite This Work

Climate Change Extended Dataset EDA (1940–2040). Analysis performed using Python (Pandas, Seaborn, Matplotlib, SciPy). Future projections (2026–2040) are model-based simulations for educational use only.

Authoritative Sources for Verification

NASA Climate: <https://climate.nasa.gov> | NOAA: <https://www.noaa.gov> | IPCC Reports: <https://www.ipcc.ch>

9. Conclusion

This Exploratory Data Analysis of the Global Climate Change Dataset (1940–2040) has examined 100 years of climate dynamics across six regions, four scenarios, and 13 variables. The findings are unambiguous in their direction, consistent across every analytical lens applied, and demand serious attention from researchers, policymakers, and institutions alike.

9.1 What the Data Tells Us

The most fundamental conclusion from this analysis is that global climate conditions are deteriorating on every measurable dimension — and they are doing so with increasing speed. Temperature anomaly, sea level rise, methane concentration, and extreme event frequency all follow upward trajectories that are tightly correlated with the passage of time ($r = 0.91\text{--}0.94$). There is no plateau, no reversal, and no indication of slowdown in any baseline trend.

The scenario comparison is where the data delivers its most important message. The difference between the BAU and Green scenarios is not marginal — it is the difference between a manageable and an unmanageable future. Green scenario outcomes are consistently the best across every single climate variable. The technology and policy frameworks to achieve Green outcomes are known and available. The gap between knowing and doing is the defining challenge of the era this dataset covers.

A counterintuitive but important finding is the structural independence of CO₂ emissions from GDP and population. Emissions are not the automatic byproduct of national wealth or size — they are the product of energy choices, industrial structure, and policy environment. This means the pathway to emission reduction runs through sectoral reform and energy transition, not through capping economic growth or constraining population. It is a message of possibility, not restriction.

9.2 What the Data Demands

Urgency over gradualism

Every year of delay locks in additional warming, sea level rise, and methane accumulation. The 1.5°C threshold will be breached by 2030–2035 under current trajectories. Incremental action is no longer sufficient.

Methane as a co-equal priority

The near-perfect correlation between methane and sea level (0.99) means any climate framework that focuses solely on CO₂ is structurally incomplete. Methane must receive equal regulatory attention and investment.

Equity-centred climate finance

Asia experiences 11 times more extreme climate events than Oceania yet is not the highest per-region emitter. Climate financing and adaptation support must be allocated proportionally to exposure, not historical contribution.

Policy and renewables as a single system

The 0.85 correlation between climate policy strength and renewable energy adoption confirms they are not independent levers. Any policy framework that separates them will underperform compared to one that advances both simultaneously.

9.3 Closing Statement

Climate change is frequently framed as a future problem. This dataset, spanning 1940 to 2040, makes clear that it has been a present and measurable reality for decades. The historical record shows consistent deterioration. The future projections show that the trajectory only worsens without decisive structural intervention. The Green scenario proves that intervention works — it produces better outcomes across every variable, in every region, across every time horizon examined.

The data does not offer false hope or false alarm. It offers the most valuable thing available in any complex challenge: an honest, evidence-based picture of where we are, how we got here, and what the consequences of different choices will be. The responsibility to act on that picture rests with every stakeholder who reads it.

Made with  for a data-driven understanding of our planet's climate future.